Lahore Garrison University

Department of Criminology

Final Year Project Proposal

Student Information:

Name:	**Email**
Contact Information:	**Supervisor:**

Project Description:

This project aims to develop MailFort AI, an intelligent, enterprise-level email forensics and threat detection system designed to automatically identify, analyze, and mitigate phishing and email-based cyberattacks within corporate environments. The solution combines Artificial Intelligence, Digital Forensics, and Threat Intelligence to scan and analyze emails, attachments, and embedded URLs in real time. It utilizes AI-based phishing intent detection, attachment hash verification via VirusTotal APIs, and blockchain-based evidence preservation to ensure forensic integrity. MailFort AI will operate as a browser extension integrated with a cloud-based DFIR dashboard, allowing corporate users and SOC analysts to detect phishing attempts, verify attachment safety, trace malicious sender origins, and generate tamper-proof forensic reports for compliance or legal purposes.

Relevance to Degree Program:

This project aligns with the BS Cyber Security and Digital Forensics curriculum by integrating cyber threat analysis, forensic evidence preservation, and AI-driven email security. It demonstrates the use of forensic automation, digital investigation techniques, and incident response methodologies in real-world corporate defense. Through MailFort AI, students will gain practical expertise in:

- Analyzing email headers and metadata for spoofing.
- Investigating malicious attachments through hash and sandbox analysis.
- Utilizing AI and NLP to detect phishing content.
- Preserving evidence integrity using blockchain-based mechanisms.

The system provides a strong bridge between academic knowledge and professional DFIR (Digital Forensics & Incident Response) practices used in the corporate cybersecurity sector.

Proposed Methodology:

- **Phase 1: Data Collection & Preprocessing**
  - Collect a dataset of phishing and legitimate emails (e.g., Enron + PhishTank datasets).
  - Extract and label fields: sender, subject, body, attachments, URLs, and header info.
  - Normalize text and convert email files (.eml/.msg) into structured JSON format.
- **Phase 2: Email Header & Metadata Forensics**
  - Parse SPF, DKIM, and DMARC validation results.
  - Analyze "Received From" chains to detect spoofed or foreign IPs.
  - Display results in a structured header inspection module.
- **Phase 3: AI-Based Phishing Detection (NLP + ML)**
  - Train AI/NLP model to detect phishing intent using transformer models (BERT or Logistic Regression).
  - Classify each email as Safe, Suspicious, or Malicious based on tone, urgency, and content anomalies.
- **Phase 4: Attachment Threat Intelligence Integration**
  - Automatically extract file hashes (MD5/SHA256) from attachments.
  - Integrate VirusTotal, Hybrid Analysis, and AbuseIPDB APIs to check malware reputation.
  - Verify file extensions and MIME types for spoofing attempts (e.g., "invoice.pdf.exe").
- **Phase 5: Browser Extension Development**
  - Build Chrome/Edge extension to automatically scan incoming emails in web clients (e.g., Gmail, Outlook).
  - Show real-time status icons (🟢 Safe / 🟠 Suspicious / 🔴 Phishing) beside each email.
  - Provide instant "Detailed Forensic Report" button for users and investigators.
- **Phase 6: DFIR Dashboard (Cloud Management Console)**
  - Develop a web-based dashboard for corporate SOCs.
  - Features: central log management, alert tracking, report generation, and multi-user collaboration.
  - Integration with SIEM platforms (Splunk, QRadar) for large-scale threat correlation.

- **Phase 7: Blockchain-Based Evidence Vault**
  - Store hashes and metadata of analyzed emails in a private blockchain ledger (Hyperledger Fabric).
  - Ensures immutability and tamper-proof forensic evidence.
- **Phase 8: Testing and Evaluation**
  - Evaluate using phishing and legitimate datasets.
  - Metrics: Precision, Recall, F1-Score, and Detection Accuracy.
  - Conduct usability testing on browser extension and dashboard interface.

Expected Outcomes:

The final system will deliver:

- A real-time phishing and email forensics browser extension for corporate users.
- An AI-powered phishing detection engine integrated with global threat intelligence sources.
- A DFIR dashboard for forensic analysis, case management, and reporting.
- A blockchain-backed evidence storage system for tamper-proof records.
- Auto-generated forensic reports (PDF/HTML) for incident documentation and compliance auditing.

Timeline:

Phase	Task	Duration
1	Dataset collection & preprocessing	2 weeks
2	Header & metadata forensics	2 weeks
3	AI phishing detection model	3 weeks
4	Attachment threat intelligence integration	3 weeks
5	Browser extension development	4 weeks
6	DFIR dashboard & blockchain vault	3 weeks
7	Testing, evaluation & documentation	2 weeks

Total Duration: ~18 weeks

Resource Requirements:

1. Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5	Intel Core i7 / Ryzen 5
RAM	8 GB	16 GB
Storage	256 GB SSD	512 GB SSD
Display	13” HD	15.6” Full HD
Internet	Required	Required

2. Software Requirements

Type	Tool	Purpose
OS	Windows 10/11, Linux	Development
Language	Python 3.9+	AI & Forensic backend
Libraries	scikit-learn, transformers, pandas	ML/NLP modules
APIs	VirusTotal, Hybrid Analysis, AbuseIPDB	Threat intelligence
Frontend	HTML, JS, CSS (Manifest V3)	Browser Extension
Backend	Flask / FastAPI	DFIR web services
DB	SQLite / PostgreSQL	Email logs & results
Blockchain	Hyperledger Fabric	Evidence integrity
s	VS Code / PyCharm	Development environment
Docs	MS Word / LaTeX	Report writing

Research Contribution:

This project bridges AI, Digital Forensics, and Corporate Cyber Defense, introducing a real-world, automated DFIR platform for phishing detection. MailFort AI contributes to research by demonstrating how machine learning, threat intelligence, and blockchain can work together to provide tamper-proof forensic evidence, improving corporate email security and incident response readiness.

Date of Submission: [Enter the date of submission]	**Signature:** [Your signature]

Supervisor Remarks:	[Accepted] / [Rejected]